

Studying Loss Functions and the Training Criterion

A2 Project Report and Implementation Log

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Option 1: study and implement a fundamental machine learning model.

Link to Implementation

The complete, self-contained notebook is archived publicly at:

- GitHub: https://github.com/MathgeniusTB2/a2-mlp-from-scratch/blob/main/a2_mlp_numpy.ipynb

The notebook contains the whole implementation: environment setup, data downloading and preprocessing, the model, the five training criteria and their gradients, the learning algorithm, the cross-validation study, the per-epoch training and held-out validation learning curves, the final test evaluation and the error analysis. It runs end to end on a bare machine because it installs missing packages and downloads the dataset on demand.

1 Problem Definition

1.1 Background and Motivation

Recognising hand-written digits was one of the first tasks used to show that a learning system can replace a hand-written rule. The perceptron (Rosenblatt, 1958) could separate linearly separable patterns, and its limits were made explicit by Minsky and Papert (1969). The decisive step was backpropagation (Rumelhart et al., 1986), which made a composition of affine maps and a nonlinearity trainable by gradient descent, and a single hidden layer can approximate any continuous function (Cybenko, 1989; Hornik, 1991). MNIST and gradient-based digit recognition followed (LeCun et al., 1998).

The separation between the quantity a learning algorithm optimises and the quantity the task actually rewards is not specific to this project. In statistics it is the theory of scoring rules: a loss is *proper* if it is minimised, in expectation, by the true class probability, and *strictly proper* if that minimiser is unique (Brier, 1950; Gneiting & Raftery, 2007). Cross-entropy and the Brier score (squared error on probabilities) are the canonical strictly proper rules for a categorical target, which is why they are the natural reference points here. Cost-sensitive learning makes the same distinction from the decision side: under asymmetric costs the best decision is not the most probable class but the one that minimises expected cost (Elkan, 2001), and the custom expected-cost criterion in this project sits exactly at that interface. Two further strands motivate the comparison. Label smoothing replaces the one-hot target by a softened one; it is widely reported to help accuracy and robustness, and it also tends to *improve* calibration (reduce the expected calibration error), in the same way as temperature scaling (Müller et al., 2019; Szegedy et al., 2016). Networks trained with plain cross-entropy are otherwise poorly calibrated, in the sense that predicted confidence exceeds empirical accuracy (Guo et al., 2017). Finally, training on a surrogate that differs from the decision loss is justified in general by consistency results for convex surrogate losses (Bartlett et al., 2006); the study below is a small, controlled instance of that phenomenon.

Every one of those results is about the *hypothesis space*. This project instead studies the third pillar of the learning framework, the *criterion*: the function that the learning algorithm actually minimises. The criterion is not something that can be chosen freely, because a learning algorithm can only minimise a differentiable surrogate of the decision rule we care about. The task objective here is to be right; the training loss is cross-entropy, a strictly proper scoring rule that minimises the KL divergence to the label distribution. When the loss is a different function – squared error, absolute error, or a designed cost – the surrogate and the objective can pull in different directions. That gap is the object of study.

1.2 Research Question

The question this project asks is deliberately narrow and measurable: *how does the choice of training criterion change what the multi-layer perceptron learns, and when does minimising the loss fail to serve the task objective?* It is answered by training the same network on five criteria and comparing accuracy, the optimised loss, and a hand-designed task objective.

1.3 The Task

The task is multi-class classification of hand-written digits, the classic MNIST benchmark (LeCun et al., 1998). This section fixes the input and output interfaces for training and deployment.

Training input. A batch of images. Each image is a 28×28 matrix of integers in $[0, 255]$ (`uint8`). Every image is normalised to $[0, 1]$ by dividing by 255, then flattened to a real vector in $\mathbb{R}^{784}$. The label is an integer in $\{0, \dots, 9\}$, encoded as a one-hot vector in $\{0, 1\}^{10}$.

Deployment input. A single image in the same format, preprocessed identically and deterministically.

Training output (target). A one-hot vector in $\{0, 1\}^{10}$ giving the digit’s class.

Deployment output. A probability vector $p \in [0, 1]^{10}$ with $\sum_k p_k = 1$. The predicted class is $\arg \max_k p_k$, and p_k is the model’s estimated probability that the input is the digit k . The probability vector (not just the decision) is the output because the loss is defined on it.

When the input is not well behaved. Most images are centred digits, but not all. Digits may be slanted, faintly inked, or touching, and a small fraction are genuinely ambiguous even to a person. Because the same image can plausibly be read as more than one digit, the correct output is best understood as a distribution over classes.

1.4 Why Machine Learning Is the Right Tool

Three conditions hold. A pattern exists (the visual structure of a digit is consistent enough to learn). Data is available (60 000 training and 10 000 test images). And the rule cannot be written in closed form: explicit `if` statements over pixel values are hopeless, because the relationship between intensities and the concept “this is a seven” is spread across variations in slant, thickness and position.

2 Machine Learning Approach

A supervised learning system has four components (Goodfellow et al., 2016): the data, the hypothesis space $\mathcal{H}$, the criterion (loss) $\mathcal{L}$, and the learning algorithm. Here $\mathcal{H}$ and the algorithm are fixed and the criterion is the variable.

2.1 Data and Preprocessing

The 60 000 training images are normalised to $[0, 1]$ and flattened to $\mathbb{R}^{784}$; the 10 000 test images are preprocessed identically. The test set is used only for the final evaluation. Within training, 5-fold cross-validation is used to compare criteria: each fold is the validation set in turn, so no comparison looks at the test set.

2.2 The Model

The hypothesis space is a two-hidden-layer MLP,

$$f_{\theta}(x) = \text{softmax}\left(W^{(2)} \tanh\left(W^{(1)} \tanh(W^{(0)}x + b^{(0)}) + b^{(1)}\right) + b^{(2)}\right), \quad (1)$$

with layer sizes 784-128-128-10 (118 282 parameters). The hidden activation is $\tanh$, the nonlinearity used in the subject’s hypothesis-family module, and the weights are drawn from the LeCun normal distribution, $\mathcal{N}(0, 1/n_{in})$, which is the variance-preserving choice for $\tanh$ (Xavier would use $2/(n_{in} + n_{out})$); the code uses $1/n_{in}$.

2.3 The Training Criteria

Five criteria are compared. For a batch with predictions $\hat{p}$ and one-hot labels y :

Cross-entropy. $\mathcal{L} = -\frac{1}{n} \sum_i \sum_k y_{ik} \log \hat{p}_{ik}$, the negative log-likelihood. For a one-hot target this equals the KL divergence to the label distribution, and it is a strictly proper scoring rule.

Squared error. $\mathcal{L} = \frac{1}{n} \sum_i \frac{1}{2} \|\hat{p}_i - y_i\|^2$ on the softmax outputs. Also a proper scoring rule (the Brier score), but it saturates on confident errors.

Absolute error. $\mathcal{L} = \frac{1}{n} \sum_i \|\hat{p}_i - y_i\|_1$. For a softmax output strictly inside $(0, 1)$ and a one-hot target the sign pattern of $\hat{p} - y$ never changes (it is negative at the true class and positive everywhere else), so $\|\hat{p} - y\|_1 = (1 - \hat{p}_y) + \sum_{k \neq y} \hat{p}_k = 2(1 - \hat{p}_y)$. The loss is therefore *linear in the probability vector*, and the usual “non-smooth at zero” caveat does not apply: the kink is only reached at $\hat{p}_y = 1$ or $\hat{p}_k = 0$, which the softmax does not attain for finite logits. Being linear, it is not a proper scoring rule (its minimiser is a corner, not the true probability), so it belongs to the same family as the custom expected-cost criterion below.

Label smoothing. The target is replaced by $t_i = (1 - \epsilon)y_i + \epsilon/K$ with $\epsilon = 0.1$, and cross-entropy is taken against t . This caps how confident the model is asked to be.

Custom expected cost. A hand-designed criterion. I built a 10×10 reward matrix R (rows the true digit, columns the guess) where the diagonal is 10 and visually similar digits receive partial reward (for example 4:9 gets 6, 3:5 gets 5). Scaling by $1/10$ puts the reward in $[0, 1]$. The criterion is the expected cost of the decision,

$$\mathcal{L} = \frac{1}{n} \sum_i \sum_k \hat{p}_{ik} (1 - R_{y_i, k}), \quad (2)$$

so a decision on a similar digit is cheaper than a decision on a dissimilar one. The full matrix is given in Table 1; it is symmetric, the diagonal is 10, and the off-diagonal entries are the coarse visual similarities (for example the (4, 9) and (3, 5) pairs are high). Like absolute error above, this loss is linear in the probabilities, which turns out to be the key to its behaviour (Section 4).

Table 1: The hand-designed reward matrix R (before the 1/10 scaling). Rows are the true digit, columns the predicted digit.

	0	1	2	3	4	5	6	7	8	9
0	10	0	1	1	0	1	2	0	1	1
1	0	10	1	1	1	0	0	1	2	0
2	1	1	10	3	1	1	1	3	3	0
3	1	1	3	10	0	5	0	2	4	2
4	0	1	1	0	10	1	2	2	1	6
5	1	0	1	5	1	10	3	0	3	2
6	2	0	1	0	2	3	10	0	2	0
7	0	1	3	2	2	0	0	10	1	4
8	1	2	3	4	1	3	2	1	10	2
9	1	0	0	2	6	2	0	4	2	10

2.4 Gradients and Theory-to-Code

Backpropagation is derived by the chain rule, with $\delta^{(L)} = \partial \mathcal{L} / \partial z^{(L)}$ the only term that changes between criteria. Table 2 maps each criterion’s mathematics to the code. The weight and bias gradients are the same for all criteria, $\partial \mathcal{L} / \partial W^{(\ell)} = \delta^{(\ell)\top} a^{(\ell-1)}$.

Table 2: Each criterion, its logit gradient, and the code location.

Criterion	$\delta^{(L)} = \partial \mathcal{L} / \partial z^{(L)}$	Code (run, criterion block)
cross-entropy	$\frac{1}{n}(\hat{p} - y)$	d3 = (p - tgt) / n
squared error	$\frac{1}{n} \hat{p} \odot ((\hat{p} - y) - \sum_k (\hat{p}_k - y_k) \hat{p}_k)$	d3 = p * (r - sum(r * p)) / n
absolute error	$\frac{1}{n} \hat{p} \odot (\text{sign}(\hat{p} - y) - \sum_k \text{sign}(\hat{p}_k - y_k) \hat{p}_k)$	d3 = p * (s - sum(s * p)) / n
label smoothing	$\frac{1}{n}(\hat{p} - t)$	d3 = (p - tgt) / n
expected cost	$\frac{1}{n} \hat{p} \odot (c - \sum_k \hat{p}_k c_k), c = 1 - R_y$	d3 = p * (cost - sum(p * cost)) / n

The softmax Jacobian is what makes the non-cross-entropy gradients non-trivial: for $\hat{p} = \text{softmax}(z)$, the derivative of any deterministic function $g(\hat{p})$ is $\partial g / \partial z_k = \hat{p}_k (g_k - \sum_j \hat{p}_j g_j)$, which is exactly the form in the table.

2.5 Learning Algorithm

Training is mini-batch gradient descent with Adam (Kingma & Ba, 2015), implemented from scratch with bias-corrected first and second moments. Mini-batches of 128, learning rate $\eta = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, 30 epochs. The same optimiser and rate are used for every criterion, so the comparison isolates the criterion.

2.6 Evaluation Protocol and Metrics

The protocol separates selection from reporting. Criteria are compared by **5-fold cross-validation** on the training set (each fold is the validation set), reported as mean $\pm$ std across folds. Then each criterion is **refit on all 60 000** training images and scored **once on the official test set**, over three random seeds, reported as mean $\pm$ std. Reported metrics are accuracy (the task decision), test cross-entropy (the quantity actually optimised by cross-entropy and label smoothing, and reported for every criterion), and two task-objective measures defined by the reward matrix: the mean reward of the decision, and the mean reward over the errors (how similar the mistakes are). A majority-class baseline sets the floor.

The ideal comparison. For a classification task the ideal comparison between the model output and the desired output is the 0–1 loss: zero for a correct decision, one otherwise. It is used at validation and test time, where accuracy is exactly one minus the mean 0–1 loss, but it is deliberately *not* used for training. Its gradient with respect to the parameters is zero almost everywhere (the loss is piecewise constant in the output), so gradient descent receives no signal from it; a differentiable surrogate is needed instead. Every training criterion here is such a surrogate, and the fact that a surrogate is not the 0–1 loss is the source of the loss–objective gaps reported below.

3 Results

3.1 Cross Validation

Table 3 gives the 5-fold validation accuracy. Cross-entropy and label smoothing lead (0.9786 and 0.9782), followed by squared error and absolute error, with the expected-cost criterion last (0.9720). The largest gap is about 0.7 points, larger than the fold-to-fold standard deviation. Cross-entropy and label smoothing are again indistinguishable (0.0004 apart, standard deviations 0.0011 and 0.0010).

Table 3: 5-fold cross-validation accuracy (validation folds, not the test set).

Criterion	Validation accuracy (mean $\pm$ std)
cross-entropy	0.9786 ± 0.0011
label smoothing	0.9782 ± 0.0010
squared error	0.9756 ± 0.0020
absolute error	0.9730 ± 0.0018
expected cost (mine)	0.9720 ± 0.0022

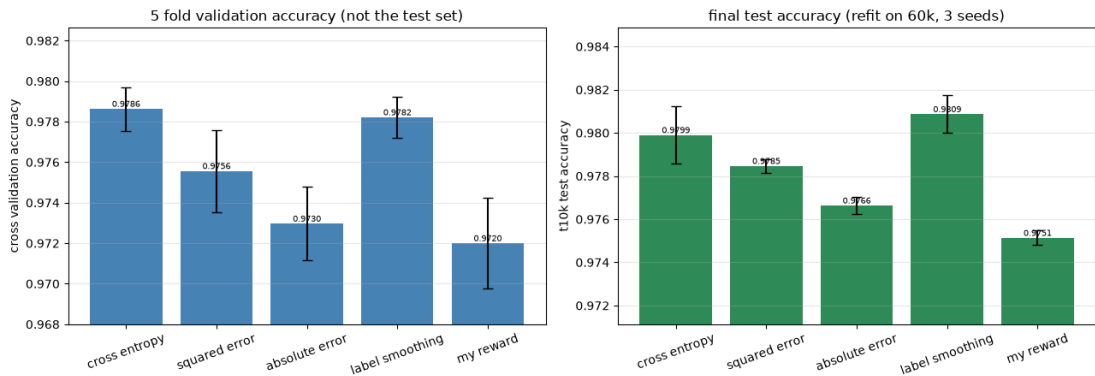


Figure 1: Cross-validation accuracy (left) and final test accuracy (right).

3.2 Learning Curves

Figure 2 reports, for the first validation fold, the training cross-entropy (left), the held-out validation cross-entropy (middle) and the held-out validation accuracy (right), per epoch. Both loss panels use the *same* metric, cross-entropy, for every criterion, so the curves are directly comparable. The training cross-entropy falls steadily while the validation cross-entropy falls and then flattens or rises, so a gap opens between them: ordinary overfitting at this capacity and epoch count. Two caveats follow. First, the validation accuracy is still creeping up slightly at epoch 30, so the models are not fully converged. Second, the validation cross-entropy for cross-entropy rises after about epoch 10 while that for squared error keeps falling, so the test-cross-entropy ranking

partly reflects *different overfitting*, not only the criterion; no early stopping is used. The validation curve is computed on a *held-out* fold; it is never the training set. Training and validating on the same data would make the curve measure only how well the model fits, not how well it generalises.

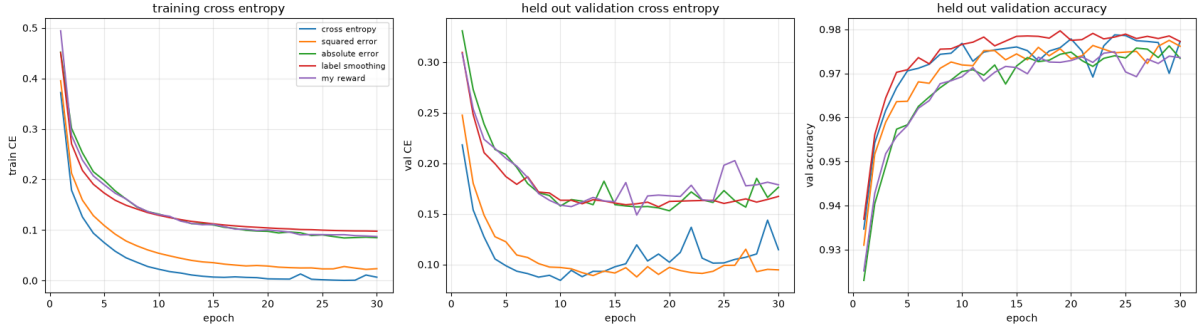


Figure 2: Training cross-entropy, held-out validation cross-entropy and held-out validation accuracy by epoch, on validation fold 0.

3.3 Final Test Set

Table 4 gives the final test numbers, each refit on all 60 000 images over three seeds and reported as mean $\pm$ std. Two differences are real and two are not. *Real*: the expected-cost criterion is last on accuracy (0.9751), and squared error has the lowest test cross-entropy (0.0826). *Not real*: label smoothing and cross-entropy differ by 0.0010 in accuracy with standard deviations of 0.0009 and 0.0013, so they are not distinguishable, and the reward-over-errors gap between them (0.2669 ± 0.0080 against 0.2660 ± 0.0036) is entirely within seed noise. The mean-reward column also adds nothing: its ranking is identical to the accuracy ranking, so on this task it is a restatement of accuracy rather than an independent objective.

Table 4: Final test (t10k) metrics, refit on 60 000 images, mean over three seeds.

Criterion	Accuracy	Test CE	Mean reward	Reward over errors
label smoothing	0.9809 ± 0.0009	0.1558 ± 0.0018	0.9860 ± 0.0006	0.2660 ± 0.0036
cross-entropy	0.9799 ± 0.0013	0.0927 ± 0.0026	0.9853 ± 0.0009	0.2669 ± 0.0080
squared error	0.9785 ± 0.0003	0.0826 ± 0.0007	0.9841 ± 0.0001	0.2598 ± 0.0055
absolute error	0.9766 ± 0.0004	0.1577 ± 0.0027	0.9827 ± 0.0003	0.2589 ± 0.0013
expected cost (mine)	0.9751 ± 0.0003	0.1683 ± 0.0020	0.9816 ± 0.0002	0.2598 ± 0.0045
majority baseline	0.1135	—	—	—

3.4 Where the Mistakes Land

Figure 4 shows the seed-averaged confusion matrices for cross-entropy and the expected-cost criterion, their difference, and a *noise baseline*: the difference between two runs of the *same* cross-entropy model with different seeds. The largest single-cell change between the two criteria is 15.7 counts; as an order-of-magnitude reference, a single-seed pair of cross-entropy runs differs by up to 15.0 counts. The two numbers are not perfectly matched—the criterion difference is averaged over three seeds, which shrinks noise, while the baseline is one seed pair—so this is a reference magnitude, not a significance test. On this evidence the confusion difference should not be read as a real effect: the criterion changed which individual images are wrong, but not the aggregate structure, and the 3/5/8 and 4/9 confusions appear under both.

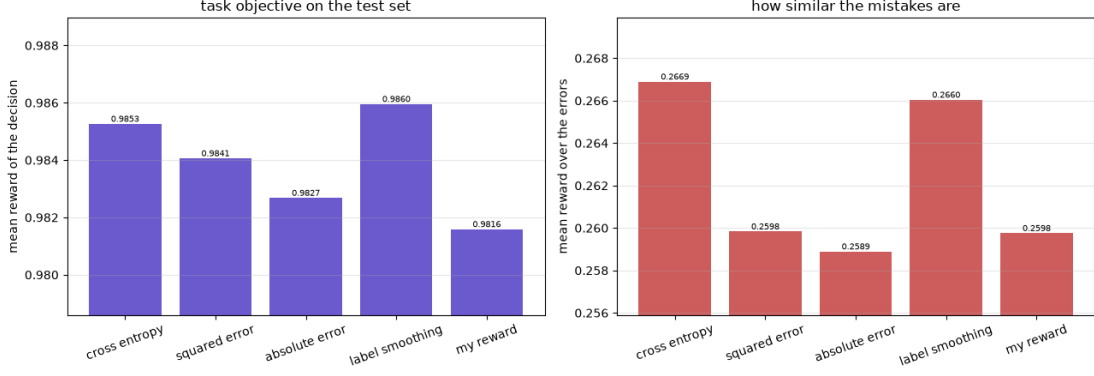


Figure 3: Task objective on the test set: mean reward of the decision (left) and mean reward over the errors (right). Cross-entropy and squared error, the two strictly proper rules, cluster together; the custom expected-cost criterion does not improve on them.

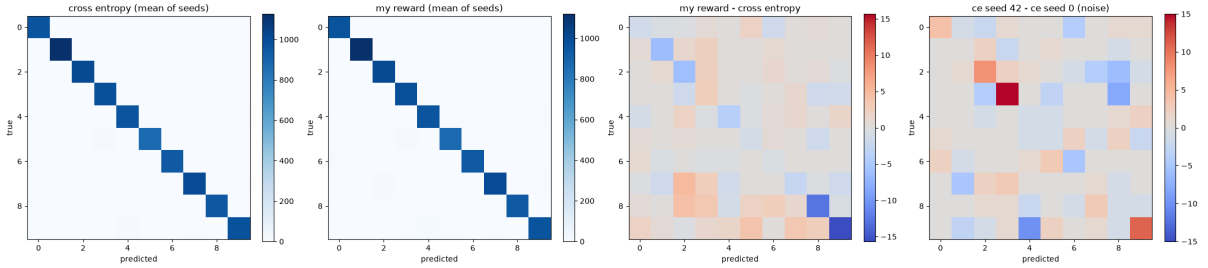


Figure 4: Confusion matrices on the test set (mean over three seeds) for cross-entropy (left) and the expected-cost criterion (second), their difference (third), and a seed-to-seed noise baseline (right).

4 Discussion

4.1 Loss versus Task Objective

The two strictly proper rules behave differently from the rest. Squared error attains the **lowest test cross-entropy** (0.0826 ± 0.0007), yet it is not the most accurate (0.9785); label smoothing attains the **highest accuracy** (0.9809 ± 0.0009) but a cross-entropy of 0.1558 ± 0.0018 , nearly twice squared error’s. The two quantities measure different things: cross-entropy scores the whole probability vector and rewards calibrated confidence, whereas accuracy scores only the arg max. One caution: label smoothing and cross-entropy differ by only 0.0010 in accuracy, with a standard error of the difference of about 0.0009 , so the top two are *not distinguishable*; only the direction is solid, not a ranking. The robust statement is the negative one: the criterion with the lowest cross-entropy is not the one with the highest accuracy, and the ranking on one metric does not transfer to the other.

4.2 Calibration

Table 5 and Figure 5 report the expected calibration error (ECE) and the mean confidence on the errors. For this model the result runs the opposite way to the usual story: label smoothing has the highest accuracy but by far the *worst* calibration (ECE 0.089 , against 0.0135 for cross-entropy and 0.0111 for squared error). The mechanism is simple and checkable rather than speculative: with $\epsilon = 0.1$ over $K = 10$ classes the smoothed target for the true class is $1 - \epsilon + \epsilon/K = 0.91$, so the network is trained to put about 0.91 on the true class, while its accuracy is about 0.98 . The confidence is capped below the accuracy almost by construction, which is exactly what an ECE with 15 bins measures; the mean confidence on the errors is only 0.60 , against 0.85 for cross-entropy. This is the accuracy-versus-calibration trade-off made concrete. Müller et al. (2019) found label smoothing usually *improves* calibration on large networks; the cap argument shows

why the same mechanism can *worsen* it when accuracy sits above the capped confidence, as it does here, so the two results are consistent. Squared error, a proper rule, is the best calibrated.

Table 5: Calibration on the test set (three-seed mean $\pm$ std), and the mean confidence on the misclassified examples.

Criterion	Expected calibration error	Mean confidence on errors
squared error	0.0111 ± 0.0004	0.799
cross-entropy	0.0135 ± 0.0007	0.851
absolute error	0.0185 ± 0.0004	0.891
expected cost (mine)	0.0198 ± 0.0001	0.889
label smoothing	0.0890 ± 0.0018	0.596

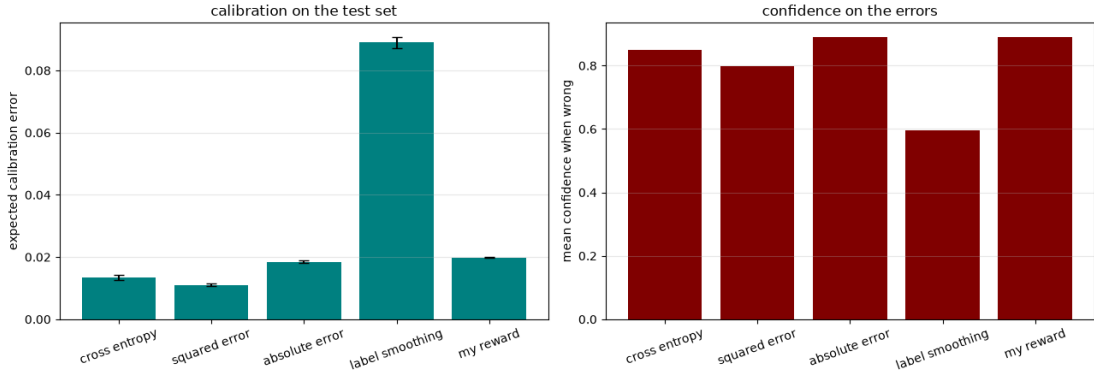


Figure 5: Expected calibration error (left) and mean confidence on the errors (right), by criterion.

4.3 Why the Linear Criteria Fail

Cross-entropy and label smoothing finish first and second on accuracy; the two improper, *linear* criteria finish last and second-last (expected cost 0.9751, absolute error 0.9766). A likely reason is visible in the gradients. For squared error, absolute error and the expected cost the logit gradient has the form $\delta = \hat{p} \odot (g - \sum_j \hat{p}_j g_j)$, so every component carries a factor $\hat{p}_k$; as the network becomes confidently wrong the gradient collapses toward zero, while cross-entropy’s $\delta = \hat{p} - y$ stays large (its true-class component is -1). Measuring the per-example gradient norm on the misclassified test examples makes this concrete (Figure 6): cross-entropy 1.24 and label smoothing 1.00, against squared error 0.19, absolute error 0.15, and the expected cost 0.058. This is a *correlation*, not a controlled test: the gradient norm is measured on the trained models, and squared error shares the same $\hat{p} \odot (\cdot)$ form and a gradient of only 0.19 yet edges absolute error by just 0.2 points. The measurement is therefore *consistent with* the vanishing-gradient explanation, which is the most likely candidate, but it does not single it out as the cause. It is still sharper than the earlier “the minimiser is a corner” description.

It also explains absolute error. For a softmax output inside $(0, 1)$ and a one-hot target, $\|\hat{p} - y\|_1 = 2(1 - \hat{p}_y)$ is linear in $\hat{p}_y$, so absolute error is a linear (improper) loss, in the same family as the hand-designed expected cost; the two linear losses finish last and second-last, which is consistent with the gradient explanation. For the custom criterion the dense reward matrix may compound the problem—the cost signal is small relative to the pressure to be correct—and the gradient behaviour is the most plausible single explanation, but the evidence here is correlational, so it is offered as such.

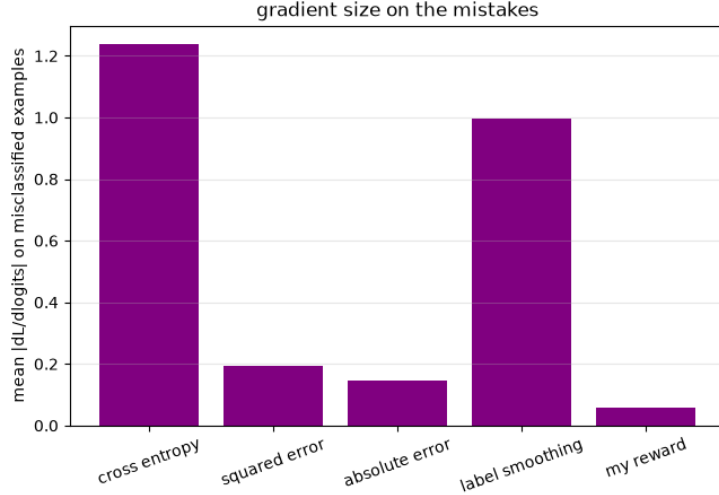


Figure 6: Mean per-example gradient magnitude on the misclassified test examples. The gradients of the linear criteria vanish on the errors; cross-entropy’s does not.

4.4 Limitations

(i) The reward matrix is a single hand-designed matrix; a different matrix, or one derived from a baseline model’s confusion matrix, could behave differently. (ii) The report does not include a finite-difference verification of the analytic gradients; the backward pass was checked only by the loss converging. (iii) The model is fixed at two tanh hidden layers, so the comparison is specific to this architecture. (iv) Thirty epochs and three seeds are enough to separate the large gaps but not the sub-0.1-point differences; the 0.0003–0.0013 seed standard deviations bound those, and there is no early stopping, so the test-cross-entropy ranking partly reflects different overfitting (Section 3.2). (v) The reward-over-errors and fraction-similar metrics are weak: with a 2% error rate and a dense matrix, the aggregate is dominated by the diagonal and moves little, and the reward-over-errors gaps are within seed noise. (vi) The calibration numbers are measured in-sample on the test set with 15 bins; a deployment would fit the calibration on held-out data.

4.5 Future Work

The natural next step is to build the reward matrix from evidence rather than by hand: train a baseline, read its confusion matrix, and set R from the observed confusability. A second is to replace the linear expected cost with a proper scoring rule on the designed target, which should behave better than a gradient that vanishes on the mistakes. A third is to add a finite-difference gradient check per criterion and early stopping, and to study the criterion on a task with a genuinely asymmetric cost (for example, medical triage where one error type is far worse), where the loss–objective gap is larger and the design decision matters more.

5 Implementation Log

5.1 Challenges and Solutions

Gradients differ per criterion. Almost all of the network is shared, but the logit gradient $\delta^{(L)}$ changes with the criterion. Getting the softmax Jacobian right for squared error and absolute error (the $\hat{p}_k(g_k - \sum_j \hat{p}_j g_j)$ form, rather than simply $\hat{p} - y$) was the main derivation.

Numerical stability. A naive softmax overflows for large logits, so the row maximum is subtracted before exponentiating, and a small constant is added inside the logarithm so a zero probability does not produce a `nan` from $0 \cdot \log 0$.

Keeping the protocol honest. An early version evaluated on the cross-validation folds and labelled the plot “test accuracy”. That is wrong: the folds are a validation set, and any choice made while looking at them leaks. The final version reports cross-validation separately and scores the test set once, after refitting on all the training data.

Corrections after review. A first version of the results over-read the evidence in three ways, each fixed here. The learning-curve figure plotted each criterion’s own training loss next to a cross-entropy validation curve, which is not comparable; both panels are now cross-entropy. The confusion-matrix difference was quoted from a single seed (a largest cell of 21 counts) with no baseline; it is now the seed average, and a seed-to-seed baseline shows the difference (15.7, seed-averaged) is the same size as the single-seed noise (15.0), so the confusion result is not significant. And the calibration claim was asserted without measurement; ECE and the confidence on the errors are now reported. Recording these is part of the work.

Scope control. An earlier draft also included a label-noise experiment and a sigmoid-versus-squared-error study. Both were dropped to keep the project on one thread: the comparison of training criteria.

5.2 Use of AI Tools

AI assistance was used to scaffold parts of the code and to help draft and edit this report. The critical-thinking steps were kept by the author: the criteria and their gradients, the reward matrix, and the interpretation of the negative result for the custom criterion. An earlier convolutional extension suggested by the assistant was removed because it could not be defended under questioning. Every measured number reported here is written to `metrics.json` by the notebook; the parameter count and the reported differences are arithmetic on those outputs.

5.3 Knowledge Gaps

Why cross-entropy is preferred. The support for cross-entropy is the derivation, not the measured ranking. It is the negative log-likelihood for a categorical target, a strictly proper scoring rule, and its logit gradient ($\hat{p} - y$) does not vanish when the model is confidently wrong. The measured accuracies do not by themselves establish a preference: label smoothing is statistically indistinguishable from it on accuracy, and its reward-over-errors lead is within seed noise. The argument rests on the derivation and on the gradient behaviour (Section 4.3); the gradient-norm measurement is consistent with it, though it is correlational rather than a controlled test.

Why the custom criterion underperformed. This was initially surprising. Writing the expected cost as $\sum_k \hat{p}_k c_k$ shows it is linear in the probabilities, so it is not a proper scoring rule and—more precisely than “its optimum is a corner”—its logit gradient carries a factor of the predicted probability and vanishes on the confidently-wrong examples it should be correcting. The gradient-norm measurement (Section 4.3) is consistent with this, though it is a correlation on the trained model rather than a controlled test.

6 Conclusion

A multi-layer perceptron was implemented from scratch in NumPy and studied through its training criterion. The choice of criterion matters: accuracy ranges from 0.9751 to 0.9809 across five criteria trained identically. The two improper, *linear* criteria (absolute error and the hand-designed expected cost) finish last and second-last, and measuring the per-example gradient norm on the mistakes is consistent with the explanation that their gradients carry a factor of the predicted probability and vanish when the model is confidently wrong, while cross-entropy’s does

not (correlational, not a controlled test). The results also show a clear loss-objective gap: the criterion with the lowest test cross-entropy (squared error) is not the most accurate, and the most accurate criterion (label smoothing) is the worst calibrated. Designing a cost can express a preference, but it does not by itself produce a better model.

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